

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		Activation energy	1			1		
		(ii)		Curve drawn with a lower activation energy under existing curve – energy state at beginning and end must be the same.	1			1		
	(b)			{Alcohol dehydrogenase/ enzyme} has a <u>specific</u> {shaped active site/ tertiary structure/ OWTTE} (1) {Ethanol/ substrate} has a <u>complementary</u> (shape) (1) (The two fit together) to form an <u>enzyme-substrate complex</u> (1)	3			3		
	(c)	(i)		Any answer between 0.57 - 0.63 = 2 marks If incorrect award 1 mark for sight of : attempted calculation of gradient		2		2	2	
		(ii)		1. P: the rate of reaction is higher as the concentration of {ethanol/ substrate} is high (1) 2. The concentration of {alcohol dehydrogenase/ enzyme} is limiting the rate of reaction; (1) 3. Q: the rate of reaction is lower the concentration of {ethanol/ substrate} decreases(1) 4. The concentration of {ethanol/ substrate} becomes the limiting factor. (1)		2	2	4		